Response to Referee Comments

Manuscript rsif-2026-0404

Perturbation-library decomposition of ECIS micromotion signals using finite-difference impedance simulations

E. Acerbo, M. I. Bellotti, F. J. Bonetto

Journal of the Royal Society Interface

October 14, 2026

We thank the Senior Editor and both referees for their thorough evaluation of our manuscript. The comments have led to substantial improvements in both the analysis and the presentation. In the following we address each comment in turn, indicating the corresponding changes made to the revised manuscript. Referee comments are reproduced verbatim in gray boxes; our responses follow immediately below each comment.

Summary of changes

The principal modifications to the revised manuscript are:

- Terminology revised: "non-degenerate" replaced by "model-consistent" throughout; "identification" removed; biological interpretations explicitly framed as model-consistent inferences.
- Giaever–Keese model description corrected: "infinite electrode" replaced by translational symmetry formulation. Eq. (2) clarified: Z_n in Eq. (1) defined as the CPE impedance alone; R_{constr} noted as external to the model, added only for experimental comparison.
- Coupling parameter K_{RB} equation added.
- Data processing section: moving-average detrending justified over linear detrending; sub-minute micromotion scope stated; excitation frequency (1 kHz) distinguished from sampling interval; sequential spectral–micromotion measurement protocol clarified.
- Library B trimmed from 90 to 36 modes (excluded $K_{RB} = 3$, arm length limited to 50% of cell side). This trimming was prompted by the reviewers' phase-space coverage questions, which led us to identify a plotting inconsistency (single-electrode library vectors compared against two-electrode experimental data). Correcting this inconsistency revealed that the largest Library B modes were never selected by the reconstruction. The trimming improved greedy RMSE for confluent ($54 \rightarrow 34\,\Omega$ in δR) and post-healing ($37 \rightarrow 16\,\Omega$ in δR) without affecting fractal dimensions.
- Library D figure (Fig. 6) right panel replaced with cube-edges visualization showing the individual parameter contributions to the phase-space trajectory.
- Library D collective nature and Library C pixel-level consistency explicitly clarified.

- All reconstructions (greedy and Viterbi, four electrodes) rerun with trimmed Library B. RMSE table (Table 2) and fractal comparison table (Table 3) updated.
- Phase-space coverage figure (Fig. 7) corrected for the two-electrode factor and regenerated with trimmed Library B; hull coverage now above 99.9% for all culture states.
- Reconstruction figure (Fig. 8) regenerated with repositioned insets and bottom-center legend.
- Mode-state figure (Fig. 9) regenerated; internal electrode identifiers replaced by culture-state labels.
- Greedy reconstruction order dependence acknowledged: permutation analysis over all 24 library orderings reported; B-first justified on phase-space grounds.
- Viterbi sequential nature clarified; computational infeasibility of joint optimization stated; editing artifact removed.
- Fractal dimension comparison reframed as consistency check, not validation. Three-level evidence framework (signal reconstruction, statistical preservation, biological validation) adopted.
- Conclusion rewritten: biological interpretation per culture scenario added; pharmacological validation proposed as future work. Closing paragraph added framing the work as a proof of concept for deterministic decomposition of ECIS micromotion, with multi-frequency dual-protocol measurements identified as the path toward unique mode identification.
- Sensitivity of transition frequency band corrected (1–3 kHz window) and electrode/cell-type dependency noted.
- Library A figure (Fig. 3) trimmed to one cell-spacing period; color and marker-size gradients widened.
- Baseline figure (Fig. 2) annotations made consistent (both relative %).

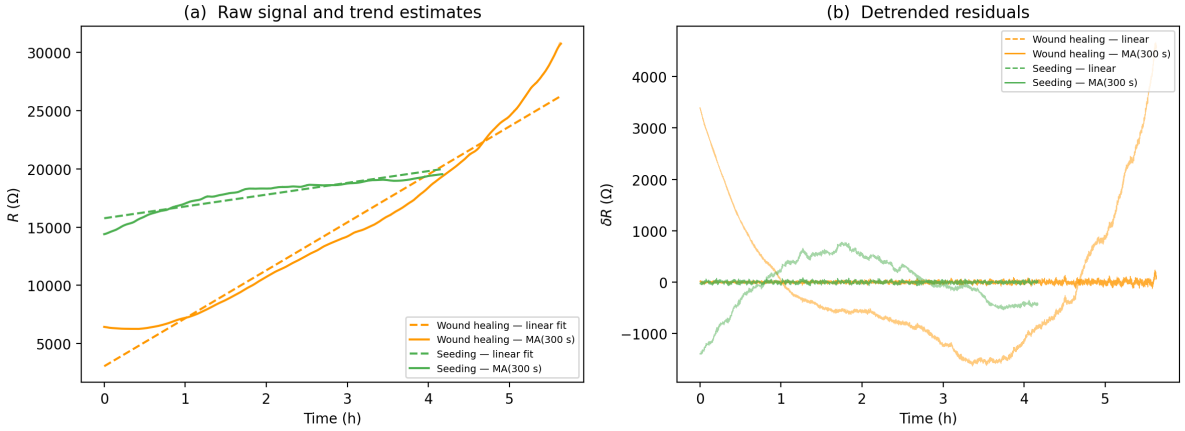
Referee 1

The Referee considers the manuscript "highly publishable" and recommends publication after some concerns have been addressed. We are grateful for the positive assessment and address each point below.

R1.1 — Linear detrending vs. moving average

> *Why not using linear detrending, i.e. subtracting a regression line instead of using a moving average? This would extend transmission of the high pass filter.*

We considered linear detrending as the reviewer suggests. However, the impedance drift in the non-stationary recordings (wound-healing and seeding) is strongly non-linear: the wound-healing signal follows a sigmoid recovery curve over several hours, while the seeding signal exhibits a monotonic attachment transient. A global regression line cannot track these non-linear trends and leaves systematic, correlated residuals that contaminate the micromotion signal. The accompanying figure illustrates this for both recordings: the linear residual retains a large-amplitude sigmoid artifact (several thousand ohms for wound-healing), whereas the moving-average residual remains centred around zero, preserving only the fast fluctuations of interest.



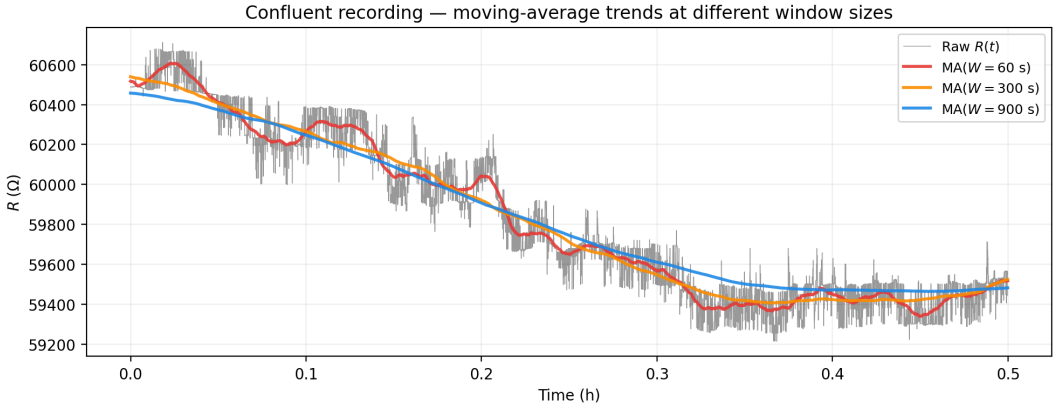
The culture evolution studied here spans several hours, during which the impedance baseline undergoes non-linear drift associated with cell attachment, spreading, or wound closure. Over these timescales, linear detrending is suboptimal as shown. The moving-average filter with window W acts as a local high-pass filter that adapts to the curvature of the baseline drift. For the two stable recordings (confluent, post-healing) the drift is small and approximately linear, so both methods produce equivalent results; in those cases we retain the moving average for methodological consistency across all electrodes. Linear detrending could be appropriate for shorter recordings, such as those obtained with a dual-instrument setup that measures spectral impedance and micromotion simultaneously, where the acquisition window is sufficiently short for the drift to remain approximately linear. A sentence clarifying this choice has been added to the Data Processing section of the revised manuscript.

R1.2 — Window size $W = 900\,\mathrm{s}$

> *Window W size 900 s (= 15 min) seems very big. Please argue why such a big window is necessary. It removes cellular noise with time constants in the minute regime.*

We note that the moving-average detrending acts as a high-pass filter: the residual $\delta R(t) = R(t) - \mathrm{MA}_W[R](t)$ retains all fluctuations at timescales shorter than W. A larger window therefore preserves more slow content rather than removing it. The accompanying figure illustrates this for a 30-minute segment of the confluent recording: the MA($W = 60\,\mathrm{s}$) trend closely tracks every minute-regime undulation, meaning those fluctuations would be subtracted from the signal and lost. The MA($W = 300\,\mathrm{s}$) still captures some multi-minute

structure. Only the MA($W = 900\,\mathrm{s}$) produces a smooth, slowly varying baseline that preserves the minute-regime fluctuations intact in the residual.



The perturbation libraries constructed in this work model collective micromotion phenomena occurring at sub-minute timescales, where the additive small-perturbation hypothesis holds. Both chosen windows ($W = 300\,\mathrm{s}$ for seeding and wound-healing, $W = 900\,\mathrm{s}$ for confluent and post-healing) comfortably preserve the entire sub-minute regime. The larger $W = 900\,\mathrm{s}$ is used for the two stable-monolayer recordings where the slow drift is small and predominantly instrumental; the shorter $W = 300\,\mathrm{s}$ is necessary for the non-stationary recordings to track the steep non-linear impedance rise and prevent drift leakage into the residual (see R1.1). Longer-timescale phenomena such as macroscopic culture reorganization or gradual monolayer maturation are not addressed by the current perturbation libraries and fall outside the scope of this first work. The manuscript text has been revised to clarify this scope. We remain open to continuing this discussion should there be further questions.

R1.3 — Infinite electrode in GK model description

> *Electrode and cell layer are not considered infinite in the Giaever–Keese model. Proof: The GK-model uses the constriction resistance which is not defined for infinite electrodes.*

We agree that describing the electrode as "infinite" was an oversimplification in our presentation. The constriction resistance indeed arises from the finite electrode geometry, as the reviewer correctly points out. However, the constriction resistance is not part of the GK cell-covered electrode model itself: in standard ECIS practice, R_{constr} is determined independently by fitting the bare electrode spectrum, subtracted from the real part of the confluent impedance prior to GK fitting, and added back only for display purposes. The specific bare electrode impedance Z_n that enters Eq. (1) is therefore the CPE impedance alone, without R_{constr}.

The analytical basis of the GK model is that every cell on the electrode is surrounded by identical neighbors, so that the current balance can be solved for a single representative cell and the result integrated over the electrode area. This translational symmetry of the homogeneous, fully confluent monolayer is the operative assumption, not an infinite electrode. The revised manuscript replaces the original sentence with:

"The Giaever–Keese (GK) model derives the specific impedance of a cell-covered electrode under the assumption that every cell on the electrode is surrounded by identical neighbors, so that the radially symmetric current balance for a single representative cell can be integrated over the electrode area."

This formulation avoids the misleading "infinite" terminology while accurately describing the model's foundation.

R1.4 — Eq. (2): R_{constr} is not part of Z_{CPE}

> *It is just minor but Eq. (2) says $Z_{\mathrm{CPE}} = \ldots$. The constriction resistance R_{constr} is not part of the CPE impedance. The serial arrangement of Z_{CPE} and R_{constr} describes the total impedance of the cell-free, bare electrode.*

We agree with the reviewer. The constriction resistance is a separate physical element in series with the CPE, not part of it. The original Eq. (2) conflated both into a single expression labelled Z_{CPE}, which is incorrect. In the revised manuscript, Z_n in Eq. (1) is defined as the CPE impedance alone:

$$Z_n = Z_{\mathrm{CPE}} = \frac{1}{A\,(j\omega)^n}. \tag{1}$$

The constriction resistance R_{constr} is determined independently from the bare electrode spectrum and added in series only when comparing to experimental data (Fig. 2); it does not enter the GK model or the simulation engine.

R1.5 — Figure 2: relative magnitude vs. absolute phase

> *Figure 2 magnitude/phase: The difference at 1 kHz is given in relative values for the magnitude but absolute values for the phase. Any reason for this? For the phase plot: Does the delta values refer to "finite difference vs GK" as well as in magnitude plot?*

Both annotations now use relative differences. The magnitude annotation was already expressed as a percentage; the phase annotation has been changed from an absolute difference in degrees to a relative percentage as well. Both annotations now read "FD vs GK: $\Delta|Z| = X.X\%$ @ 1 kHz" and "FD vs GK: $\Delta\varphi = X.X\%$ @ 1 kHz", making the convention consistent across the two panels. The figure has been regenerated accordingly.

R1.6 — Square cells vs. GK disk cells

> *The GK-model treats cells as disks with a circular foot print. The simulations presented in the manuscript treat cells as squares. Is there any conversion needed or integrated?*

No geometric conversion is needed. The finite-difference engine discretises the electrode–cell system on a square-pixel grid and solves the current balance numerically without assuming

circular symmetry. The square cell geometry was designed to have the same cell area a_c as the GK disk, and the parametric sensitivity structure is conserved: independent perturbations of α, R_b, and C_m in the finite-difference model produce relative impedance variations that closely follow the corresponding GK analytical variations, as demonstrated in the figure attached below (used in a companion manuscript currently under review at Physical Review E.).

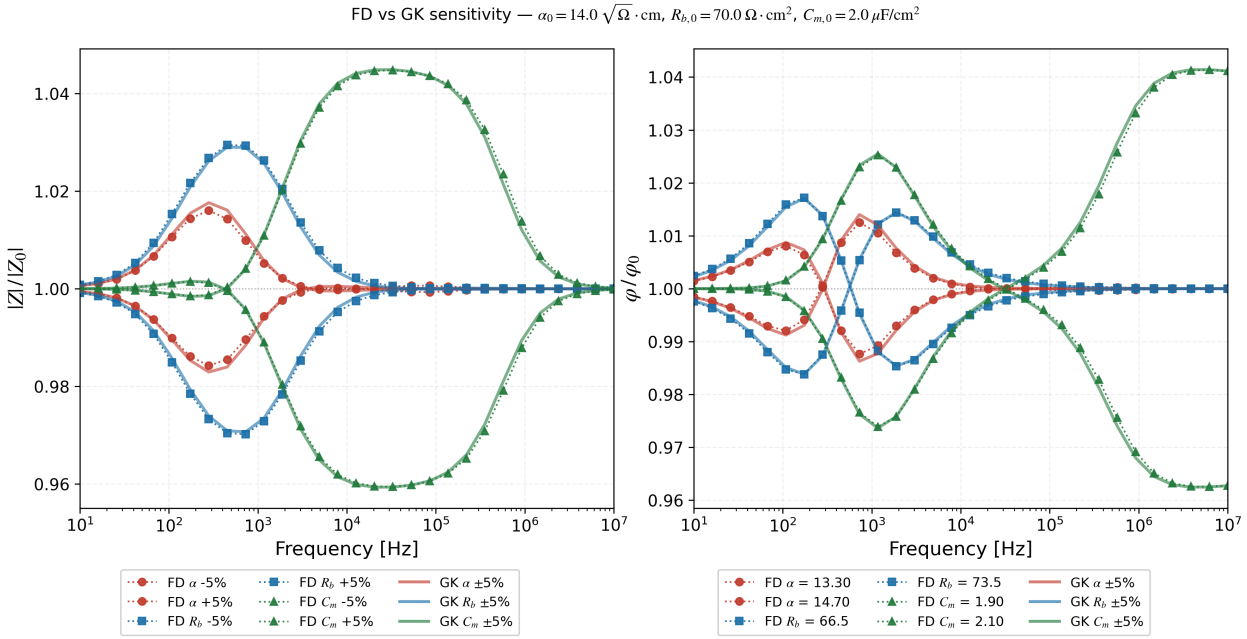


Figure 1: Relative impedance variation using the finite difference model by varying each biophysical parameter in a range of ±5% against a reference set $Z_{FD}(\alpha = 14\ \sqrt{\Omega}\cdot\mathrm{cm},\ R_b = 70\ \Omega\cdot\mathrm{cm}^2,\ C_m = 2\times10^{-6}\ \mathrm{F/cm}^2)$. Red curves correspond to the variation of α, blue to R_b and green to C_m. Solid lines show the corresponding GK model variation normalised to the same reference parameter set.

The impedance estimated by the finite-difference model is not identical to the analytical GK prediction, as shown in Fig. 2 of the manuscript, where both models are evaluated with the same parameter set and compared to experimental confluent data. The small residual differences arise from features that the numerical model resolves but the analytical model does not: current migration from the electrode to the insulating substrate through the cleft space, the square cell footprint versus the cylindrical GK cell, and the finite pixel width of tight junctions (versus infinitesimally thin in the analytical model). Nevertheless, the agreement is within 3% in $|Z|$ and 0.7° in phase at 1 kHz.

R1.7 — Library D: collective or individual perturbations?

Libraries A, B and C represent collective translation of the monolayer grid relative to the electrode, collective rearrangement or collective tight junction widening across the cell layer. Library D represents perturbations that produce impedance fluctuations with-

out geometric rearrangements. But are the latter also collective? Or are they individual and randomly distributed across the cells within the monolayer. To account for individual, non-collective movement of individual cells seems important to catch the nature of micromotion.

All four libraries, including Library D, apply collective perturbations: every cell on the electrode receives the same fractional parameter change simultaneously. In this framework the perturbed α, R_b, and C_m values should be interpreted as effective culture-wide values rather than as properties of any individual cell.

We agree with the reviewer that individual, spatially heterogeneous cell movements are an important aspect of micromotion. The simulation engine can in principle assign different parameter values to each sub-cellular node, enabling the modelling of individual cell variations. However, this was not pursued in the present work because the number of possible configurations grows combinatorially and would not proportionally increase the phase-space coverage in the two-dimensional (δR, δX) observation space available at a single frequency. A spatially heterogeneous perturbation library, potentially guided by statistical distributions of cell-level parameter values, is a natural extension for future work.

The manuscript has been revised to state explicitly that Library D perturbations are collective, and a note on the absence of individual perturbation modes has been added to the Discussion.

R1.8 — Library C: geometry change affects current paths

Library C: The authors state that there is a collective geometric re-arrangement from square to brick-wall without altering individual cell properties? But when you change geometry, current paths change which will affect model parameters R_b and α. Is this consistent?

The reviewer's concern is physically valid: changing the geometric arrangement of cells does alter the current paths and therefore the macroscopic impedance. This is precisely what Library C captures. The simulation input parameters (α, R_b, C_m) are global values that define the impedance of each sub-cellular pixel node, and in doing so the cellular impedance. When alternating columns are displaced, each cell footprint remains identical and every pixel retains the same membrane or junction conductance. What changes is the relative position of neighboring cell membranes, which modifies the current-path topology across the electrode. The finite-difference engine re-solves the full current balance on the rearranged pixel geometry, so the resulting impedance change is a direct consequence of the altered current distribution, not of any parameter modification.

In other words, R_b and α as simulation inputs (pixel-level conductances) are unchanged, but the effective R_b and α that one would infer from fitting the GK analytical model to the resulting spectrum do change, because the analytical model assumes a fixed packing. This distinction between input parameters for the numerical model and effective analytical

parameters is a natural consequence as the cell order can vary in the former and with it the estimated spectrum.

R1.9 — Figure 3: line color and marker size not visible

> *Figure 3: Please specify the meaning of "line color and marker size increase with displacement magnitude". The reviewer does not see any "color increase" and no marker size increase.*

We acknowledge that in the submitted figure the color and marker-size gradients were too subtle to distinguish, compounded by the large number of overlapping curves. The original figure plotted all displacements up to two cell spacings (40 pixels), at which point the curves loop back due to the translational symmetry of the square grid, producing visually identical overlapping traces.

The figure has been revised. In the left panel (perturbation spectrum), the displacement range has been trimmed to half a cell-spacing period (10 pixels, $\approx$10 µm), eliminating redundant symmetric curves and reducing visual clutter. The color gradient has been widened and the transparency now scales with displacement, making individual curves clearly distinguishable. The right panel (boundary check) shows the full displacement range (0–40 µm), displaying the periodic structure over two cell spacings. The caption has been updated to reflect these changes.

R1.10 — Definition of coupling parameter K_{RB}

> *Please provide a definition of the coupling parameter K_{RB}. In the current manuscript it has not been specified.*

We agree that the definition was missing. The coupling parameter K_{RB} rescales the tight-junction resistance as a function of the geometric arm length:

$$R_b(l_w,\, K_{RB}) \;=\; R_{b,0}\left(1 + K_{RB}\, l_w / l_{w,\max}\right),$$

where $R_{b,0}$ is the baseline junction resistance, l_w is the L-corner arm length in pixels, and $l_{w,\max}$ is the maximum interior arm length. The physical motivation is that when intercellular mechanical tension deforms the junction geometry, the effective barrier resistance changes in proportion to the deformation extent. Because the fraction of cells deforming in a common direction in the real monolayer is unknown, multiple coupling values were tested to explore the resulting range of perturbation directions in the $(\delta R,\, \delta X)$ plane. The equation and the definition of all terms have been added to the revised manuscript.

R1.11 — Single-frequency identifiability: 2 observables vs. 3 parameters

A general comment. The authors measure the complex impedance at a single frequency. The measurement provides two measurands $|Z|$ and phase. How can you unequivocally trace back from single freq impedance measurements to three parameters of the model (R_b, α, C_m)? Does not seem plausible. Or in other words: a combination of $|Z|$ and phase may have more than one solution for R_b, α, C_m even within the parameter windows $\pm 1\%$.

The reviewer raises a fundamental and valid point. The reconstruction does not perform a parameter inversion: it does not attempt to recover a unique (α, R_b, C_m) triplet from the two observables $(\delta R, \delta X)$ at a single frequency. Instead, it selects the nearest pre-computed library mode in the two-dimensional $(\delta R, \delta X)$ plane. Multiple modes, potentially from different libraries, can map to nearby points in this plane, and the decomposition is therefore not guaranteed to be unique.

We have revised the manuscript to replace the term "non-degenerate" with "model-consistent" throughout, reflecting that the reconstruction identifies the best-matching perturbation mode within the library rather than a uniquely determined physical state. The four-library sequential structure partially mitigates the ambiguity, because each library occupies a characteristic direction in the $(\delta R, \delta X)$ plane (see R1.13), but does not eliminate it entirely.

It is worth noting that in the experimental protocol, a full spectral impedance sweep is acquired prior to each micromotion recording, ensuring correct electrode functioning and culture state. The spectral measurement provides a GK fit of the confluent monolayer, which determines the reference biophysical parameter set (α, R_b, C_m) used as the baseline for library construction and for the spectral comparison in Fig. 2. No spectral information is available during the micromotion recording itself. This spectral anchoring constrains the parameter regime in which the reconstruction operates, reducing the effective degeneracy compared to a purely unconstrained search. We have clarified this protocol in the revised manuscript.

The natural next step to further resolve the degeneracy is to measure spectral impedance and micromotion simultaneously, using a dual-instrument setup with two lock-in amplifiers and two function generators, so that the reconstruction can operate in a higher-dimensional observation space at each time step.

A quantitative analysis of the identifiability, including nearest-neighbor distances between library modes and the sensitivity of the reconstruction to alternative mode assignments, is presented in our response to Referee 2, Major concern 1 (R2-M1), where the same issue is addressed in detail.

R1.12 — Role of C_m: fixed or varied?

How do paracellular barrier properties or cell-substrate cleft geometry affect C_m? Or was C_m kept constant across all calculations which would make the whole approach more comprehensible.

In both the GK analytical model and the numerical finite-difference model, C_m is a membrane property (the area-specific impedance of the apical and basal membranes in series) and is independent of paracellular barrier properties (R_b) or cleft geometry (α). Accordingly, C_m is held constant across all calculations in Libraries A, B, and C, where only geometric or junction modifications are applied.

C_m is varied only in Library D, where it is perturbed by $\pm 1\%$ alongside α and R_b. The justification for including C_m perturbations is that junction-related changes in the cell, which are not explicitly resolved in the cell footprint geometry, can modify the effective cell surface area and therefore the area-specific membrane impedance. For instance, membrane ruffling, or bleb formation alter the effective apical surface without changing the cell footprint, producing a measurable shift in C_m. Including these perturbations in Library D allows the reconstruction to capture such membrane-level fluctuations alongside cleft and junction changes.

R1.13 — Figure 6 right panel incomprehensible

> *Figure 6, right panel: the reviewer does not understand what is presented. The authors must spend a few more explanations to make this graph comprehensible or simplify it.*

We acknowledge that the high point density in the scatter plot hinders interpretation for readers encountering it for the first time. The objective of this panel is to show that the combined $\pm 1\%$ perturbations of α, R_b, and C_m span all four quadrants of the (δR, δX) phase space, providing the broadest directional versatility among the four libraries. In the original scatter, only the R_b value determines the color, as encoding the remaining two parameters via marker shape or size would add confusion without being informative given the point density.

The right panel of Fig. 6 in the revised manuscript has been replaced with a cube-edges visualization showing only the 12 edges of the perturbation cube, where two of the three parameters are fixed at their extreme values ($\pm 1\%$) and the third is swept. This reveals the individual contribution of each parameter to the phase-space trajectory: α perturbations (red) produce the longest excursions, R_b (blue) contributes primarily along the diagonal, and C_m (green) acts nearly horizontally. The 8 cube corners (black diamonds) mark the simultaneous $\pm 1\%$ extremes.

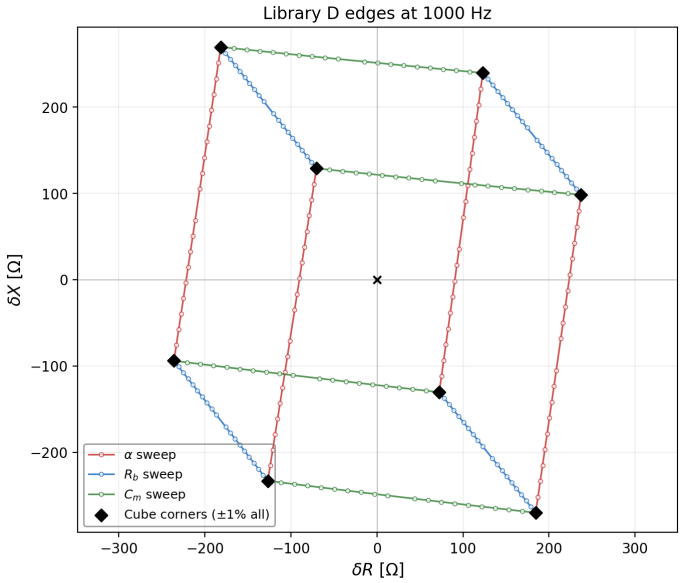


Figure 2: Revised right panel of Fig. 6. Phase-space representation of Library D at 1 kHz, showing only the 12 edges of the perturbation cube. Along each edge, two of the three biophysical parameters are held at their extreme values (±1%) while the third is swept through the full range. Red: α sweep (R_b and C_m fixed); blue: R_b sweep (α and C_m fixed); green: C_m sweep (α and R_b fixed). Black diamonds mark the 8 cube corners where all three parameters are simultaneously at ±1%.

The manuscript caption has been expanded to explain how the three parameters contribute to the phase-space coverage.

R1.14 — Libraries computed for confluent only

> *The libraries were computed for confluent cultures only. Is it meaningful to apply them to the other scenarios at all as the latter are very different from the confluent phenotype?*

We acknowledge that this is a first-order approximation. The manuscript already discusses this limitation explicitly in the Conclusions (calibration factor paragraph): at low confluence, individual cells behave as isolated particles rather than coupled elements of a collective, and the micromotion dynamics may be qualitatively different from those captured by the confluent-baseline libraries. Designing libraries for non-confluent states requires a fundamentally different approach: partially covered electrodes admit many possible spatial arrangements of cells, each producing a different impedance response. This demands an ensemble of simulations over randomized cell configurations rather than the single deterministic geometry used for the confluent case, substantially increasing the computational cost and the library design complexity. Extending the library construction to include partially covered electrode configurations is identified as future work.

R1.15 — Figure 8: insets obscure color codes

Figure 8: Place the inserts differently to make color codes visible.

We thank the reviewer for pointing this out. The residual insets have been replaced by a dedicated third column of residual panels, eliminating occlusion entirely. The legends have been reformatted to a single-column layout for compactness. The figure has been regenerated.

R1.16 — Transition frequency wording in Conclusion

In the conclusion, it must read: "below this transition frequency, the measurement. . . ".

We acknowledge the error. The original sentence incorrectly placed the sensitivity description relative to the transition frequency. The revised text now describes a sensitivity window: the perturbation amplitude peaks in a frequency band approximately between 1 kHz and 3 kHz for most libraries, below which the CPE electrode response dominates and above which the constriction resistance dominates, both reducing the sensitivity to cellular perturbations. The sentence has been corrected in the revised manuscript.

R1.17 — Transition frequency depends on electrode and cell type

The transition frequency is dependent on the type of electrode and the cell type. In the conclusion, this dependency is neglected.

The reviewer is correct. The sensitivity band identified in the manuscript is specific to the MDCKII / 350 µm electrode combination and the associated biophysical parameters used here. Different electrode geometries (e.g., larger 96W10E+ arrays or interdigitated electrodes), electrolyte interfaces, or cell types with different (α, R_b, C_m) values would shift this frequency band. A sentence stating this dependency has been added to the Conclusion.

R1.18 — Conclusion lacks cell-biological impact

Conclusion: I was hoping to get a clear-cut answer what mechanisms drives the micro-motion for a given cell type in a given experimental scenario. Is it cell substrate adhesion or barrier function? Or is it the membrane? This particular aspect has not been met with clarity. I would recommend a much more feet-on-the-ground conclusion what is the cell biological impact of the modelling, what has been learned about cell behavior from these calculations. To me that is a bit missing. The space is there as the current conclusion reads more like a summary and could be shortened significantly not to be repetitive.

We agree that the original conclusion was too focused on methodology and did not sufficiently address what was learned about cell behavior. The revised conclusion has been shortened by condensing the algorithmic summary into a single paragraph and replacing the repetitive methodological content with a new paragraph that interprets the reconstruction results in terms of cell biology for each culture scenario:

- **Confluent:** Libraries B and D dominate the reconstruction, indicating that the micromotion of a mature monolayer is driven primarily by fluctuations in tight-junction geometry and small-amplitude biophysical parameter drift, consistent with homeostatic maintenance of barrier function.
- **Seeding:** Library A occupancy broadens relative to the confluent case, indicating a larger contribution from lateral cell displacement. This is consistent with cells that have not yet established firm substrate adhesion and are able to slide on the electrode surface.
- **Wound healing:** Library D shows an anticorrelation between $\delta\alpha$ and $(\delta R_b, \delta C_m)$, suggesting that cells simultaneously tighten the cell-substrate cleft while stiffening paracellular junctions. This signature is consistent with cytoskeletal contractility during active migration into the wound area.
- **Post-healing:** The narrowest mode occupancy across all libraries and the lowest fractal dimension indicate that the monolayer has reached a mechanically quiescent state with reduced micromotion complexity following wound closure and radial cell reordering.

These interpretations are model-consistent inferences from the decomposition, not independently validated biological conclusions (see also R2-M4). The revised conclusion now opens with these biological findings before discussing limitations and future directions.

Referee 2

We thank the Referee for the detailed and constructive review. The concerns regarding identifiability and validation are well taken and have led to significant additions to the manuscript.

Major concerns

R2-M1 — Identifiability of the decomposition

The most fundamental issue is whether the proposed decomposition is actually identifiable from the available ECIS measurements. At 1 kHz, each time point is represented by only two observables, δR and δX, whereas the reconstruction simultaneously assigns states from four libraries. Consequently, many combinations of perturbations can potentially produce very similar or even nearly identical points in the $(\delta R, \delta X)$ plane. The statement that the reconstruction provides a "non-degenerate" physical description therefore appears stronger than what is demonstrated experimentally.

The authors should provide a quantitative analysis of the uniqueness/identifiability of the reconstructed states. For example, they could report the nearest-neighbor distances between library modes, the number of experimentally indistinguishable combinations, sensitivity/Jacobian analyses, or reconstruction uncertainty associated with alternative library combinations. It would be particularly important to demonstrate whether substantially

different combinations of Libraries A–D can produce comparable RMSE values. Without such an analysis, the reconstructed parameters should perhaps be described as model-consistent representations rather than uniquely identified biological mechanisms.

We agree that the identifiability question is central and that the original "non-degenerate" characterisation was overstated. The term has been replaced by "model-consistent" throughout the revised manuscript.

To quantify the degeneracy we performed a nearest-neighbor analysis across all four libraries at 1 kHz. The table below reports the within-library nearest-neighbor (NN) distances, which measure how densely each library samples the (δR, δX) plane:

Library	Modes	Min NN (Ω)	Median NN (Ω)	Max NN (Ω)
A (shift)	861	0.00	0.00	0.00
A (unique δZ)	55	0.40	3.12	8.68
B (junction)	36	16.7	288.2	801.2
D (parameter)	9261	3.9	6.2	18.2
C (brickwall)	21	0.0	22.2	43.1

Library A shows zero nearest-neighbor distances because the periodic cell spacing (20 pixels) and the square-grid mirror symmetries create multiple shifts with identical impedance responses. When these symmetry-equivalent modes are collapsed, Library A reduces to 55 unique impedance values with a median NN distance of 3.12 Ω, comparable to Library D. This geometric degeneracy does not affect reconstruction quality, as all equivalent shifts produce the same δZ.

Library B occupies a unique region of the phase space with wide inter-mode spacing (median 288 Ω), making its modes clearly distinguishable from all other libraries (0% cross-library overlap). In contrast, Library D is densely sampled and fully encompasses the phase-space footprints of Libraries A and C: 100% of A modes and 100% of C modes have a D-mode neighbour within 6 Ω. This confirms that D provides a superset of the perturbation directions available in A and C, which is one of the reasons A and C are placed last in the reconstruction sequence (see R2-M2).

For the experimental confluent recording (59 811 time steps), we queried the 50 nearest library modes for each data point and assessed cross-library ambiguity using a tolerance factor ϵ: for each time step, the best-matching mode has Euclidean distance d_{best} in (δR, δX), and all modes within $\epsilon \times d_{\text{best}}$ are counted as alternatives.

Tolerance ϵ	Mean alternatives	Median distinct libraries	Single-library (%)
1.5×	7.9	1	92.3
2.0×	11.1	1	88.6
5.0×	23.6	1	82.5

At 1.5× the best-match distance, 92.3% of time steps have all alternative modes within the

same library, meaning the cross-library ambiguity is limited. The alternative modes predominantly come from the dense Library D sampling rather than from a competing physical mechanism. Even at a conservative 5× tolerance, 82.5% of time steps remain single-library.

These results support describing the decomposition as model-consistent: at each time step, the library assignment is dominated by a single mechanism, though multiple modes within that library may produce comparable fits. A fully unique inversion would require measurements at multiple frequencies, as discussed in the Conclusions of the revised manuscript.

R2-M2 — Order sensitivity of sequential decomposition

The reconstruction algorithm processes the libraries in a fixed order: $B \to D \to A \to C$. At each stage, the selected contribution is subtracted before the next library is considered. This procedure is inherently order-dependent. A perturbation selected by Library B changes the residual presented to Library D, which subsequently changes the residual available to Libraries A and C. Thus, the statement that Libraries B and D are the "principal contributors" may partly reflect the algorithmic ordering rather than the underlying biology.

The authors should therefore perform an order-sensitivity analysis, repeating the reconstruction with different library orders and reporting how the inferred contribution distributions and RMSE change. Alternatively, a simultaneous optimization over all libraries should be considered. At minimum, the manuscript should explicitly acknowledge that the inferred relative importance of the four mechanisms is conditional on the chosen sequential decomposition.

We performed the order-sensitivity analysis requested by the reviewer, running the greedy reconstruction on the confluent recording with all 24 permutations of the four libraries. The table below reports the total RMSE for each order, sorted from best to worst:

Rank	Order	RMSE_R (Ω)	RMSE_X (Ω)	RMSE (Ω)
1	B→C→A→D	34.2	44.4	56.1
2	B→A→C→D	33.7	45.6	56.7
3	B→D→C→A	35.8	49.1	60.7
4	B→A→D→C	32.7	51.3	60.8
5	B→D→A→C	33.9	52.7	62.7
6	B→C→D→A	38.6	49.5	62.8
7	A→B→C→D	67.0	42.5	79.4
		...		
24	C→A→D→B	130.3	198.8	237.7

The results reveal a clear structure:

1. **Library B first is the determining factor.** All six top-ranked orders begin with B (RMSE 56–63 Ω), while orders that place B late produce RMSE above 220 Ω. This is

consistent with the identifiability analysis (R2-M1): B occupies a unique, well-separated region of the (δR, δX) plane with no cross-library overlap, so its contribution is unambiguously identified regardless of what follows.

2. **Among B-first orders, the spread is small.** The six B-first permutations span 56–63 Ω (10% relative spread), indicating that the relative placement of A, C, and D after B has a modest effect on the total RMSE.

3. **The statement that Libraries B and D are the principal contributors is not an artifact of ordering.** Library B dominates the reconstruction regardless of its position, and the RMSE difference between the six B-first orders is not significant.

We retain the B→D→A→C order used in the original manuscript. Although B→C→A→D achieves a marginally lower RMSE (56.1 vs 62.7 Ω), the difference is within the inter-order variability of the B-first group. The B→D→A→C sequence follows a conservative criterion of processing libraries from largest to smallest phase-space coverage, ensuring that at each stage the residual is matched against the most versatile available library. The revised manuscript now explicitly acknowledges the order dependence and reports the permutation analysis.

R2-M3 — Viterbi is sequential, not globally joint

> *The manuscript presents the Viterbi approach as an alternative temporal reconstruction. Still, the text indicates that the same sequential library structure is retained, with Viterbi optimization performed within each library after the preceding library has already been subtracted. Therefore, the Viterbi method does not appear to perform a global optimization over the combined A–D state space. It instead regularizes each library sequentially. This distinction should be made much clearer.*

The reviewer is correct. The Viterbi regularization is applied independently within each library pass on the sequential residual; it does not perform a global optimization over the joint B × D × A × C state space. A fully joint Viterbi solve would require a combined state space exceeding 10^9 modes, making the dynamic-programming matrix computationally prohibitive in both memory and time.

The revised manuscript now states this explicitly in the Viterbi description, clarifying that the temporal regularization operates per-library on the sequential residual rather than jointly across all libraries. A stray editing artifact in the submitted manuscript has also been removed.

R2-M4 — Fractal dimension agreement ≠ biological validation

> *The authors use agreement between the fractal dimensions of the experimental and reconstructed signals as evidence that the discrete representation preserves the temporal correlation structure. However, preservation of a global fractal-dimension statistic does not establish that the physical mechanisms assigned to individual fluctuations are correct.*

Many different reconstructed time series can have similar fractal dimensions while having very different temporal trajectories and physical interpretations.

Thus, I suggest that the authors distinguish clearly between:

1. *reconstruction of the impedance signal;*
2. *preservation of statistical properties of the signal; and*
3. *validation of the inferred biological/physical mechanism.*

The current analysis provides evidence primarily for (1) and partially for (2), but not yet for (3). An independent validation would substantially strengthen the manuscript. For example, perturbations that selectively alter cell–substrate adhesion, tight-junction integrity, cytoskeletal contractility, or cell movement could be used to test whether the corresponding library component changes in the predicted direction.

We agree with the reviewer's distinction and have restructured the manuscript accordingly. The revised text now explicitly separates three levels of evidence:

1. **Signal reconstruction:** the greedy and Viterbi algorithms reconstruct $\delta R(t)$ and $\delta X(t)$ with quantified RMSE. This is demonstrated directly by the reconstruction figures and RMSE values.
2. **Preservation of statistical properties:** the fractal dimension of the reconstructed signals agrees with the experimental values within windowed uncertainty. This was previously framed as "validation"; the revised manuscript reframes it as a *consistency check* on the temporal correlation structure, not as evidence for the correctness of the assigned physical mechanisms.
3. **Biological mechanism validation:** independent confirmation that a given library component changes in the predicted direction under a known perturbation. The current work does not provide this level of evidence.

The revised manuscript now states explicitly that the current analysis provides evidence for levels (1) and (2) but not yet for (3). The biological interpretations offered in the Conclusions are presented as model-consistent inferences, not as validated mechanisms.

For future validation, pharmacological perturbations that selectively target specific biophysical parameters would be informative: cytochalasin D disrupts the actin cytoskeleton and should produce a measurable shift in the Library D $\delta\alpha$ component; EGTA chelates extracellular calcium and opens tight junctions, which should be reflected in Library B and Library D δR_b changes; and wound-healing assays with known closure rates provide a reference for Library A lateral displacement trajectories. These experiments are identified as future work in the revised Conclusions.

Minor concerns

R2-m1 — Terminology: reconstruction vs. decomposition vs. identification

> *The manuscript should consistently distinguish between "reconstruction," "decomposition," and "identification." Reconstruction of δR and δX does not necessarily imply identification of the underlying biological mechanism.*

We have revised the manuscript to use these terms consistently. "Reconstruction" refers to the signal-level fitting of $\delta R(t)$ and $\delta X(t)$. "Decomposition" refers to the assignment of library modes at each time step. The term "identification" has been removed entirely, as the current work does not provide independent biological validation of the inferred mechanisms. The biological interpretations in the Conclusions are now explicitly framed as model-consistent inferences (see R1.18 and R2-M4).

R2-m2 — "Identifies dominant physical mechanisms" too strong

> *The statement that the greedy algorithm "identifies the dominant physical mechanisms" appears too strong given the current validation. A more cautious formulation would be appropriate unless independent biological validation is added.*

The phrase has been removed. The revised manuscript uses "model-consistent" in place of "non-degenerate" and frames all biological interpretations as inferences from the decomposition, not as validated mechanisms (see R1.18, R2-M1, R2-M4).

R2-m3 — Sampling interval vs. excitation frequency

> *Since the sampling time influences the sensitivity of data analysis, at 1 kHz, what is the sampling time for the micromotion measurement? (i.e., what is the time interval between each micromotion data point?) Please provide the exact sampling interval corresponding to the 1 kHz micromotion measurements and clarify whether the raw ECIS instrument output was sampled at exactly 1 kHz or measured using a 1-kHz excitation frequency. These are conceptually different quantities.*

We thank the reviewer for requesting this clarification. The 1 kHz value refers to the excitation frequency of the sinusoidal signal applied to the electrodes, not to the temporal sampling rate. The ECIS instrument uses a lock-in amplifier that demodulates the electrode voltage at the excitation frequency and outputs one impedance measurement per acquisition cycle. The lock-in time constant is the same for all recordings; the small variation in sampling interval across experiments arises from sensitivity scale verification and adjustments in the lock-in amplifier when the impedance amplitude crosses between scales:

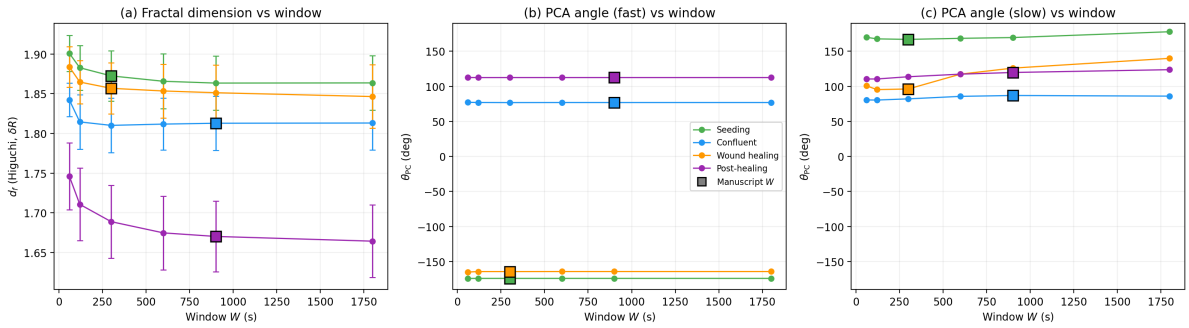
Culture state	Δt (s)	Data points	Duration (h)
Seeding	0.234	61 633	4.17
Confluent	0.266	59 811	4.68
Wound healing	0.266	76 034	5.64
Post-healing	0.188	428 928	22.02

This distinction and the sampling intervals have been added to the revised manuscript.

R2-m4 — Window sensitivity: does W affect phase-space and fractal dimensions?

> *The adaptive detrending procedure uses different moving-average windows (900 s versus 300 s) for different culture states. The authors should examine whether the choice of window affects the subsequent phase-space distributions and fractal dimensions.*

We swept the detrending window W from 60 s to 1800 s for all four electrodes and recomputed the Higuchi single-slope fractal dimension and PCA principal-axis angles (fast and slow components, with $T_{\text{cut}} = 30$ s fixed). The figure below summarises the results; square markers indicate the manuscript W values.



Three conclusions follow from this analysis:

1. **Fractal dimension plateaus above $W \approx 120$ s** for all electrodes. The difference between $W = 300$ s and $W = 900$ s is below 0.02 in d_f for every recording, well within the windowed uncertainty. The relative ordering across culture states (post-healing lowest, seeding/wound-healing highest) is preserved at all W values tested.

2. **PCA fast angles are completely insensitive to W**, as expected: the fast component (< 30 s) lies well below any detrending window in the sweep.

3. **PCA slow angles are stable for the two stable-monolayer recordings** (confluent and seeding) but show W-dependent drift for wound healing, where the steep sigmoid recovery leaks into the slow component at large W. This justifies the adaptive choice of $W = 300$ s for the non-stationary recordings.

Additionally, we computed the fraction of experimental $(\delta R, \delta X)$ points falling inside the Minkowski sum hull of the four libraries at each W. At the manuscript window values

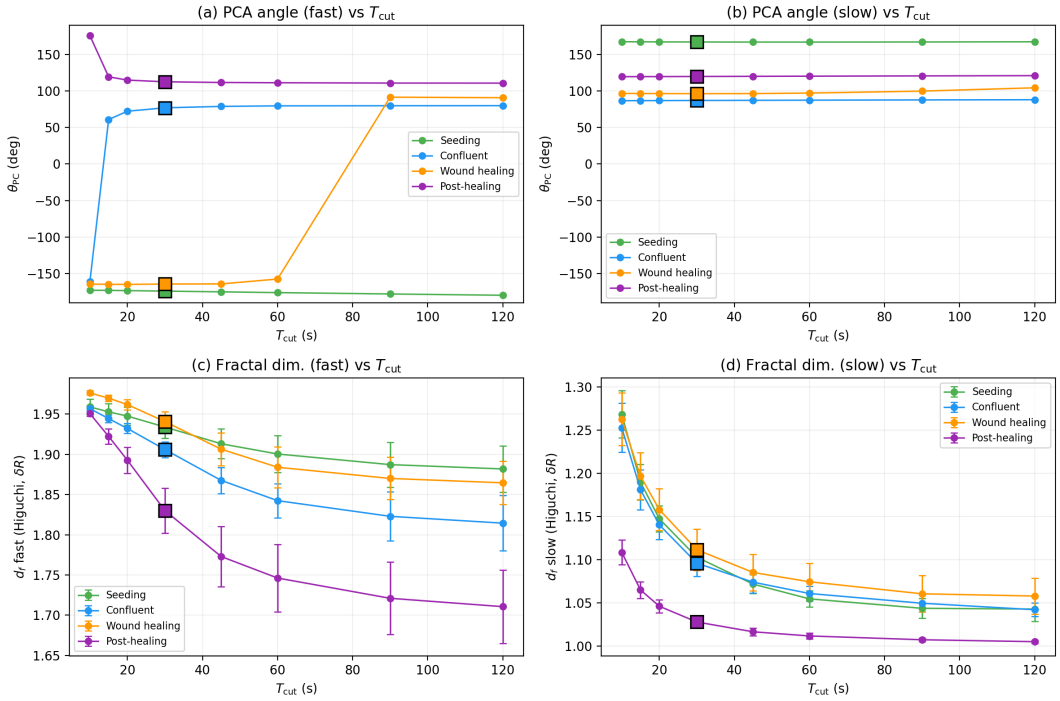
($W = 300$ s for seeding and wound healing, $W = 900$ s for confluent and post-healing), the coverage is above 99.9% for all culture states. Only at $W = 1800$ s does a measurable fraction fall outside the hull (confluent 95.7%, post-healing 99.4%), confirming that the adaptive W is well within the range where the phase-space distribution is fully covered by the perturbation libraries.

The adaptive W does not bias the fractal dimension comparison between culture states, the fast-component PCA analysis is entirely independent of the detrending window, and the phase-space coverage by the perturbation libraries is preserved at all manuscript W values.

R2-m5 — 30-s fast/slow cutoff justification and sensitivity

> *The 30-s cutoff used to separate "fast" and "slow" components appears somewhat empirical. Please justify this value and a sensitivity analysis showing whether the conclusions are robust to alternative cutoff values.*

The 30 s cutoff separates individual cell motility events (membrane ruffling, protrusion/retraction cycles, which occur on 10–30 s timescales) from slower collective reorganization. To verify that the conclusions are robust to this choice, we swept T_{cut} from 10 s to 120 s with the manuscript detrending window fixed, recomputing PCA angles and Higuchi fractal dimensions for both fast and slow components.



The analysis shows:

1. **PCA angles stabilize above** $T_{\mathrm{cut}} \approx 20$ **s** for both fast and slow components. Below 15 s, the fast angle for post-healing and confluent shows transient behaviour as the

cutoff approaches the sampling resolution. Above 20 s, all curves are flat, indicating that the qualitative conclusions (e.g., the angular separation between fast and slow in post-healing) are not sensitive to the exact cutoff value within the 20–120 s range.

2. **Fractal dimensions shift monotonically with** T_{cut}, as expected: a larger cutoff assigns more content to the slow component, making the fast residual smoother (d_f decreases) and the slow component rougher. However, the relative ordering across culture states is preserved at every T_{cut} tested. Post-healing consistently yields the lowest fractal dimension in both fast and slow components.

3. **The 30 s value sits in the transition region** between the steep initial drop and the plateau, making it the most informative choice for distinguishing culture states while remaining within the timescale range associated with individual cell motility events.
